

Lecture on Nash's embedding theorem

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The fundamental theorem of Riemannian geometry often refers to the existence and uniqueness of Levi-Civita connection. But it also refers to the result of John Nash [6] that *any Riemannian manifold is isometric to a smooth submanifold in a Euclidean space*. In particular, Riemannian manifold can be defined as a submanifold of Euclidean space with induced metric tensor.

We will prove this theorem for compact manifolds and indicate its proof in general case.

In the proof we use the so-called Günther trick — an essential simplification of the original proof that was found by Matthias Günther [2]. We also use several minor improvements found by Deane Yang, Terence Tao, and Ralph Howard [3, 8, 9] and cut corners everywhere we can.

Acknowledgments. I want to thank Fedor Nazarov for his help.

A Induced metric

Choose a smooth n -dimensional manifold Ω . Recall that a smooth field of bilinear symmetric forms, say g , on the tangent bundle $T\Omega$ is called metric tensor. Here and further, smooth means C^∞ .

Let $\bar{\Omega}$ be another smooth manifold and let $w: \Omega \rightarrow \bar{\Omega}$ be a smooth map. Recall that derivative of w in direction of a tangent vector $x \in T_p\Omega$ is denoted by xw and

$$xw = dw(x);$$

that is, the derivative of w in the direction x is the differential dw of w evaluated at x . Note that $xw \in T_{w(p)}\bar{\Omega}$.

Suppose that $\bar{\Omega}$ is equipped with a metric tensor $\bar{g}$. Consider the metric tensor g defined on Ω defined by

$$g(x, y) = \bar{g}(xw, yw).$$

It is called induced metric tensor, or pullback of $\bar{g}$ to Ω . This relation can be also written as $g = w^*\bar{g}$. Equivalently, we can say that w is an isometric map from (Ω, g) to $(\bar{\Omega}, \bar{g})$.

A metric tensor g is called Riemannian if it is positive; that is, if $g(x, x) > 0$ for any tangent vector $x \neq 0$. A smooth manifold equipped with a Riemannian metric is called Riemannian manifold.

Note that if $g = w^*\bar{g}$ is Riemannian metric, then w has to be regular; that is, the differential $d_p w$ has rank n at any point $p \in \Omega$. In particular, w is an immersion.

The Euclidean space $\mathbb{R}^q$ will be considered with standard Euclidean metric defined by the scalar product

$$g(X, Y) := \langle X, Y \rangle,$$

which is a Riemannian metric.

1. Main theorem. *Any n -dimensional Riemannian manifold (Ω, g) admits an isometric embedding into a Euclidean space $\mathbb{R}^q$ for some q that depends only on n .*

2. Exercise. *Construct a metric tensor g on the torus $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ such that $g(X, X) \geq 0$ for any tangent vector X , but there is no smooth map $w: \mathbb{T}^2 \rightarrow \mathbb{R}^{10}$ with the induced metric g .*

B Q-form

Suppose $v, w: \Omega \rightarrow \mathbb{R}^q$ are two smooth maps. Let us write $g = Q(v, w)$ if

$$g(X, Y) \equiv \frac{1}{2} \cdot [\langle Xv, Yw \rangle + \langle Xw, Yv \rangle].$$

The operator Q is symmetric and bilinear; it takes two smooth maps $v, w: \Omega \rightarrow \mathbb{R}^q$ and spits a metric tensor $g = Q(v, w)$ on Ω .

3. Observation. *A metric tensor g is induced by a smooth map $w: \Omega \rightarrow \mathbb{R}^q$ if and only if $g = Q(w, w)$.*

The components of $g = Q(w, w)$ in a local chart can be written as

$$g_{ij} = Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle.$$

4. Exercise. *Let w be a smooth map from a smooth Riemannian manifold (Ω, g) to the Euclidean space $\mathbb{R}^q$. Show that w is isometric if and only if it is length-preserving; that is, if*

$$\text{length } \gamma = \text{length}(w \circ \gamma)$$

for any curve γ in Ω .

Let Ω be a smooth manifold. Given two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{q_2}$, denote by $w = v_1 \oplus v_2$ the map

$$w: \Omega \rightarrow \mathbb{R}^{q_1+q_2} = \mathbb{R}^{q_1} \oplus \mathbb{R}^{q_2}$$

defined by $w: p \mapsto (v_1(p), v_2(p))$.

5. Exercise. *Show that*

$$Q(v_1 \oplus v_2, v_1 \oplus v_2) = Q(v_1, v_1) + Q(v_2, v_2)$$

for any two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{q_2}$.

Recall that support of a map is the closure of the set of its arguments with nonzero values.

6. Exercise. *Let $v_1, v_2: \Omega \rightarrow \mathbb{R}^q$ be smooth maps defined on a smooth manifold Ω . Suppose v_1 and v_2 have disjoint supports. Show that*

$$Q(v_1 + v_2, v_1 + v_2) = Q(v_1, v_1) + Q(v_2, v_2).$$

C Reduction to torus

Let us denote by $\mathbb{T}^n = \mathbb{R}^n / (2\pi \cdot \mathbb{Z})^n$ the n -dimensional torus.

The torus is a smooth manifold; it comes with a global system of periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$. A metric tensor on $\mathbb{T}^n$ can be also written in these coordinates; so we can (and will) think that a metric tensor is a smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$; here and further

$$N := \frac{n \cdot (n + 1)}{2}.$$

These observations make it slightly easier to work with torus, then with general manifold.

In the proof of 1 we will reduce the general case to its partial case formulated below and then prove the restricted version.

7. Restricted theorem. *Given a Riemannian metric g on the n -dimensional torus $\mathbb{T}^n$ there is an isometric immersion $w: (\mathbb{T}^n, g) \rightarrow \mathbb{R}^q$ for some q that depends only on n .*

From immersion to embedding

8. Claim. *We can assume that w in 7 is an embedding.*

Proof. Choose a Riemannian metric g on $\mathbb{T}^n$.

Choose a smooth embedding $v_0: \mathbb{T}^n \rightarrow \mathbb{R}^{2 \cdot n + 1}$; it exists by the Whitney theorem. (In fact there is a smooth embedding $v_0: \mathbb{T}^n \rightarrow \mathbb{R}^{n+1}$, but it is not going to be important for us.) Let $g_0 = Q(v_0, v_0)$; in other words, g_0 is the metric induced by v_0 . Since v_0 is an embedding, g_0 must be Riemannian.

We can assume that $g > g_0$; if not, pass to rescaling $\varepsilon \cdot v_0$ of v_0 for small $\varepsilon > 0$. In other words the metric $g_1 = g - g_0$ is Riemannian.

Applying the reduced theorem we get a smooth isometric immersion $v_1: (\mathbb{T}^n, g_1) \rightarrow \mathbb{R}^q$. Note that $w = v_0 \oplus v_1$ is an isometric embedding $(\mathbb{T}^n, g = g_0 + g_1) \rightarrow \mathbb{R}^{q+2 \cdot n+1}$. Indeed, w is isometric by 5, and it is an embedding since v_0 is. $\square$

Compact case

This part of reduction relies on the following exercise.

9. Exercise. *Let Ω be a closed submanifold in the torus $\mathbb{T}^n$. Show that for any Riemannian metric on Ω is a restriction of a Riemannian metric on $\mathbb{T}^n$.*

Reduction 1 to 7 in the compact case. Assume Ω is compact. By the Whitney embedding theorem, there is a smooth embedding $\iota: \Omega \rightarrow \mathbb{R}^{2 \cdot n+1}$. Since Ω is compact, we can assume in addition that ι is short and its image lies in a small ball; if not, then pass to the rescaling $\varepsilon \cdot \iota$ of ι for small $\varepsilon > 0$. Composing ι with the covering map $\mathbb{R}^n \rightarrow \mathbb{T}^{2 \cdot n+1}$, we get an embedding $\Omega \hookrightarrow \mathbb{T}^{2 \cdot n+1}$.

In particular, we can assume that Ω is a compact submanifold of a torus. By 9, any Riemannian metric g on Ω is the restriction of a Riemannian metric, say $\bar{g}$, on $\mathbb{T}^{2 \cdot n+1}$.

Choose an isometric smooth embedding $w: (\mathbb{T}^{2 \cdot n+1}, \bar{g}) \rightarrow \mathbb{R}^q$ provided by 8. Observe that the composition $f \circ \iota: (\Omega, g) \rightarrow \mathbb{R}^q$ is isometric. $\square$

Noncompact case

For noncompact manifold Ω , reduction 1 to 7 is more technical. It is based on the following two exercises.

10. Exercise. *Let Ω be a connected noncompact manifold. Show that it admits a smooth proper bounded embedding in $\mathbb{R}^q \setminus \{0\}$. Moreover, given a Riemannian metric g on Ω , we can assume that the induced metric of this embedding is smaller than g .*

Let Ω be an n -dimensional manifold. A smooth map $r: \Omega \rightarrow \mathbb{S}^n$ will be called chart-like if there is an open disc $D \subset \Omega$ such that the restriction $r|_D$ is a diffeomorphism $D \rightarrow \mathbb{S}^n \setminus \{p\}$ for some point $p \in \mathbb{S}^n$ and $r(x) = p$ for any $x \in \Omega \setminus D$. The disc D is called the domain of the chart-like map r .

The following exercise is a combination of partition of unity and dimension theory.

11. Exercise. Let (Ω, g) be an n -dimensional noncompact connected Riemannian manifold then there is a countable collection of chart-like maps $r_i: \Omega \rightarrow \mathbb{S}^n$ with domains D_i and a sequence of Riemannian metrics h_i on $\mathbb{S}^n$ such that

$$g = \sum_j r_j^* h_j$$

and the sequence D_i can be divided into $n+1$ subcollections such that $\bar{D}_i \cap \bar{D}_j = \emptyset$ if D_i and D_j belong to one subcollection; here $\bar{D}_i$ denotes the closure of D_i .

Reduction of the main theorem to its compact case. Let $w: \Omega \rightarrow \mathbb{R}^q$ be an embedding provided by 10 for metric $\frac{1}{2} \cdot g$. Let g_0 be the metric induced by w ; note that $g_1 = g - g_0$ is a Riemannian metric.

Let us apply 11 to (Ω, g_1) ; denote by r_i , h_i , and D_i the obtained maps, metrics and domains. Since $\mathbb{S}^n$ is compact, there is an isometric embedding $w_i: (\mathbb{S}^n, h_i) \rightarrow \mathbb{R}^q$; we can assume that $w_i \circ r_i$ maps all points in $\Omega \setminus D_i$ to zero; in particular, each w_i has compact support.

Let $S_1, \dots, S_{n+1}$ be the subdivision of natural numbers provided by 11; that is $j, k \in S_i$ for some i if and only if D_j and D_k belong to one of subcollections of domains. Set

$$s_i = \sum_{j \in S_i} w_j \circ r_j \quad \text{and} \quad s = s_1 \oplus \dots \oplus s_{n+1}.$$

By 6 and 5,

$$\begin{aligned} Q(w \oplus s, w \oplus s) &= Q(w, w) + \sum_i Q(s_i, s_i) = \\ &= g_0 + \sum_j Q(w_j \circ r_j, w_j \circ r_j) = \\ &= g_0 + \sum_j r_j^* h_j \\ &= g_0 + g_1 = g. \end{aligned}$$

Since w is an embedding, so is $w \oplus s$, hence the result. $\square$

D Free maps

As before, Ω is a smooth n -dimensional manifold.

Recall that a C^1 -map $f: \Omega \rightarrow \mathbb{R}^q$ is called regular if its differential df has rank n at each point. In other words, for any point $p \in \Omega$ and

some (and therefore any) local coordinates at p all n first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

A C^2 -map $f: \Omega \rightarrow \mathbb{R}^q$ is called free (or two-regular) if an analogous property holds for first *and second* partial derivatives; that is, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all $n + \frac{n \cdot (n+1)}{2}$ partial derivatives $\partial_i f, \partial_{ij} f$ for $i \leq j$ are linearly independent.

12. Exercise. Show that there is no free map $\mathbb{S}^2 \rightarrow \mathbb{R}^4$.

13. Exercise. Let $f_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $f_2: \Omega \rightarrow \mathbb{R}^{q_2}$ be smooth maps defined on a smooth manifold Ω . Suppose f_1 is free; show that so is $f_1 \oplus f_2$.

Let f be a smooth function on a manifold Ω . Denote by E_f the smooth map $\Omega \rightarrow \mathbb{R}^2$, defined by

$$E_f(x) = (\cos[f(x)], \sin[f(x)]).$$

In other words, E_f is the composition of two maps

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell} \mathbb{S}^1 \subset \mathbb{R}^2,$$

where $\ell: \mathbb{R} \rightarrow \mathbb{S}^1$ is the natural covering maps $y \mapsto (\cos y, \sin y)$.

14. Exercise. Consider the (x, y) -plane $\mathbb{R}^2$. Show that the map

$$E_x \oplus E_y \oplus E_{x+y}: \mathbb{R}^2 \rightarrow \mathbb{R}^6 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2$$

is free.

Generalize the statement to maps $\mathbb{R}^n \rightarrow \mathbb{R}^{n \cdot (n+1)}$.

15. Exercise. Let $f: \Omega \rightarrow \mathbb{R}^s$ be a regular smooth map and $F: \mathbb{R}^s \rightarrow \mathbb{R}^q$ is free. Show that the composition $F \circ f: \Omega \rightarrow \mathbb{R}^q$ is free.

Apply the Whitney embedding theorem and 14 to conclude that any smooth manifold admits a free embedding into a Euclidean space.

E Digression into Fourier series

Recall that $\mathbb{T}^n = \mathbb{R}^n / 2 \cdot \pi \cdot \mathbb{Z}^n$ denotes the n -dimensional torus; it comes with a standard system of periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$.

For an L^2 -function $u: \mathbb{T}^n \rightarrow \mathbb{R}^q$ the coefficients in its Fourier series

$$\sum_{m \in \mathbb{Z}^n} a_m \cdot \exp(i \cdot \langle m, x \rangle)$$

form an n -dimensional sequence from $\mathbb{C}^q$. The vector a_m for $m \in \mathbb{Z}^n$ will be denoted by $\mathcal{F}_m(u)$; it is defined as the average value

$$\mathcal{F}_m(u) := \oint_{x \in \mathbb{T}^n} u(x) \cdot \exp(-i \cdot \langle m, x \rangle),$$

here x stands for a point in $\mathbb{T}^n$ as well as a corresponding point in $\mathbb{R}^n$ defined modulo $2 \cdot \pi \cdot \mathbb{Z}^n$.

By Parseval's formula, we have

$$\|u\|_{L^2(\mathbb{T}^n, \mathbb{R}^q)} = \sqrt{\sum_{m \in \mathbb{Z}^n} |\mathcal{F}_m(u)|^2}.$$

Let us define k^{th} Sobolev norm as

$$\|u\|_{H^k(\mathbb{T}^n, \mathbb{R}^q)} = \sqrt{\sum_{m \in \mathbb{Z}^n} (1 + |m|^2)^k \cdot |\mathcal{F}_m(u)|^2}.$$

The subspace $L^2(\mathbb{T}^n, \mathbb{R}^q)$ of all functions with finite k^{th} Sobolev norm will be denoted by $H^k(\mathbb{T}^n, \mathbb{R}^q)$. It is straightforward to check that $u \in H^k(\mathbb{T}^n, \mathbb{R}^q)$ if and only if u and all its partial derivatives¹ up to order k lie in $L^2(\mathbb{T}^n, \mathbb{R}^q)$.

16. Embedding theorem. *The inclusion*

$$C^k(\mathbb{T}^n, \mathbb{R}^q) \subset H^k(\mathbb{T}^n, \mathbb{R}^q) \subset C^s(\mathbb{T}^n, \mathbb{R}^q)$$

holds for any nonnegative intergers s, k , and n such that $k > s + n/2$. In particular,

$$C^\infty(\mathbb{T}^n, \mathbb{R}^q) = \bigcap_s H^s(\mathbb{T}^n, \mathbb{R}^q).$$

If $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$, then

$$\mathcal{F}_m(\partial_j u) = i \cdot m_j \cdot \mathcal{F}_m(u).$$

Therefore,

$$\mathcal{F}_m(u - \Delta u) = (1 + |m|^2) \cdot \mathcal{F}_m(u).$$

It follows that $1 - \Delta$ defines an isometry $H^{k+2}(\mathbb{T}^2) \rightarrow H^k(\mathbb{T}^2)$; therefore,

$$(1 - \Delta)^{-1} : H^k(\mathbb{T}^2) \rightarrow H^{k+2}(\mathbb{T}^2)$$

is an isometry.

¹A partial derivative $w = \partial_j u$ is defined as a function $w \in L^2(\mathbb{T}^n, \mathbb{R}^q)$ such that equality $\int_{\mathbb{T}^n} \langle w, \varphi \rangle = \int_{\mathbb{T}^n} \langle u, \partial_j \varphi \rangle$ holds for any smooth function $\varphi: \mathbb{T}^n \rightarrow \mathbb{R}^q$.

F Formulations of key lemmas

Recall that $\mathbb{T}^n$ denotes the n -dimensional torus; it comes with a global system of periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$, and a metric tensor on $\mathbb{T}^n$ can be also written in these coordinates; so we can (and will) think that metric tensor is smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$; here and further $N := \frac{n \cdot (n+1)}{2}$.

Choose $0 < \alpha < 1$, say $\alpha = \frac{1}{2}$. Denote by $|x - y|_{\mathbb{T}^n}$ the standard distance between points $x, y \in \mathbb{T}^n$. Let us recall the definition of Hölder norms of a function $u: \mathbb{T}^n \rightarrow \mathbb{R}$

$$\begin{aligned} \|u\|_{0,\alpha} &:= \sup_{x \in \mathbb{T}^n} \{ |u(x)| \} + \sup_{x, y \in \mathbb{T}^n} \left\{ \frac{|u(x) - u(y)|}{|x - y|_{\mathbb{T}^n}^\alpha} \right\}, \\ \|u\|_{k,\alpha} &:= \max_{|\mathcal{I}| \leq k} \{ \|\partial_{\mathcal{I}} u\|_{0,\alpha} \}, \end{aligned}$$

where $\mathcal{I} = (i_1, \dots, i_n)$ denotes multi-index; so, $\partial_{\mathcal{I}} := \partial_1^{i_1} \dots \partial_n^{i_n}$ and $|\mathcal{I}| := i_1 + \dots + i_n$. The space of functions with finite norm $\|\cdot\|_{k,\alpha}$ is called the (k, α) -Hölder space; it is denoted by $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

17. Exercise. Let $v \in C^\infty(\mathbb{T}^n, \mathbb{R})$. Show that there is a constant c such that

$$\|u \cdot v\|_{k,\alpha} \leq c \cdot \|u\|_{k,\alpha}$$

for any $u \in C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

Further, let us define $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ as the space of functions $u: \mathbb{T}^n \rightarrow \mathbb{R}^q$ with all coordinate functions from $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$. Let

$$\|u\|_{k,\alpha} := \max_i \{ \|u_1\|_{k,\alpha}, \dots, \|u_q\|_{k,\alpha} \},$$

where $u_1, \dots, u_q$ are the coordinate functions of u .

From now on, let us fix a smooth free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ and let

$$L(w) := 2 \cdot Q(f, w).$$

18. Nash's lemma. The linear operator

$$L: C^\infty(\mathbb{T}^n, \mathbb{R}^q) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^N)$$

has a right inverse

$$M: C^\infty(\mathbb{T}^n, \mathbb{R}^N) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^q)$$

such that

$$\|Mh\|_{2,\alpha} \leq a \cdot \|h\|_{2,\alpha}$$

for some constant a .

19. Günther's lemma. *There is a symmetric bilinear operator*

$$\tilde{Q}: C^\infty(\mathbb{T}^n, \mathbb{R}^q) \times C^\infty(\mathbb{T}^n, \mathbb{R}^q) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^q)$$

such that

$$\begin{aligned} L\tilde{Q} &= Q, \\ \|\tilde{Q}(w, w)\|_{2,\alpha} &\leq b_2 \cdot \|w\|_{2,\alpha}^2, \\ \|\tilde{Q}(w, w)\|_{k,\alpha} &\leq b_2 \cdot \|w\|_{2,\alpha} \cdot \|w\|_{k,\alpha} + b_k \cdot \|w\|_{k-1,\alpha}^2 \end{aligned}$$

for any integer $k \geq 3$ and some fixed constants $b_2, b_3, \dots$

G Perturbation

Recall that $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ is a fixed smooth free map and

$$L(w) := 2 \cdot Q(f, w).$$

20. Perturbation lemma. *There is $r > 0$ such that the equation*

$$\textcircled{1} \quad L(w) + Q(w, w) = h$$

has a smooth solution $w: \mathbb{T}^n \rightarrow \mathbb{R}^q$ for any smooth metric h such that $\|h\|_{2,\alpha} < r$. Moreover, we can assume that

$$\|w\|_{2,\alpha} \leq C \cdot \|h\|_{2,\alpha},$$

for some constant C .

Suppose that a metric tensor g is induced by f . Note that the equation

$$Q(f + w, f + w) = g + h$$

is equivalent to $\textcircled{1}$. Indeed,

$$\begin{aligned} Q(f + w, f + w) &= Q(f, f) + 2 \cdot Q(f, w) + Q(w, w) \\ &= g + L(w) + Q(w, w). \end{aligned}$$

Therefore the perturbation lemma implies the following.

21. Corollary. *Let g be induced by a smooth free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ and h another metric tensor on $\mathbb{T}^n$. If $\|h\|_{2,\alpha}$ is sufficiently small, then $g + h$ is induced by a smooth map $\mathbb{T}^n \rightarrow \mathbb{R}^q$.*

Proof of 20. Recall that M , $\tilde{Q}$, a , and b_2 are provided by 18 and 19. Set $R = \frac{1}{10 \cdot b_2}$ and $r = \frac{R}{10 \cdot a}$. Assume $\|h\|_{2,\alpha} \leq r$. By 19,

$$\Phi: w \mapsto Mh - \tilde{Q}(w, w)$$

is a contraction map from the closed ball $\bar{B}[0, R] \subset C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ to itself.

Consider the sequence of maps $w_0, w_1, \dots: \mathbb{T}^n \rightarrow \mathbb{R}^q$ such that $w_0 = 0$ and $w_{n+1} = \Phi(w_n)$ for any n . Note that w_n is smooth for each n . Since Φ is a contraction, w_n converges in $C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ as $n \rightarrow \infty$; denote its limit by w . Further,

$$\|w\|_{2,\alpha} \leq R \quad \text{and} \quad w = Mh - \tilde{Q}(w, w).$$

Since $LM = \text{id}$ and $L\tilde{Q} = Q$, applying L to both sides of the last equation, we get that w solves ❶.

It remains to show that w is smooth. Suppose that

❷ *the sequence $\|w_n\|_{k,\alpha}$ is bounded for every integer $k \geq 2$.*

Since

$$\|w\|_{k,\alpha} \leq \lim_{n \rightarrow \infty} \|w_n\|_{k,\alpha}$$

it follows that $w \in C^{k,\alpha}$ for any k ; whence $w \in C^\infty$.

Let us prove ❷ by induction on k . The base case $k = 2$ is proved already.

By the induction hypothesis, $\|w_n\|_{k-1,\alpha}$ is bounded. In particular, there is a constant c_k such that

$$\|Mh\|_{k-1,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq c_k$$

for any n ; here b_k is from 19.

By the second estimate in 19, we get

$$\begin{aligned} \|w_{n+1}\|_{k,\alpha} &\leq \|Mh\| + b_2 \cdot \|w_n\|_{2,\alpha} \cdot \|w_n\|_{k,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq \\ &\leq c_k + b_2 \cdot R \cdot \|w_n\|_{k,\alpha} \leq \\ &\leq c_k + \frac{1}{10} \cdot \|w_n\|_{k,\alpha}. \end{aligned}$$

In particular,

$$\|w_n\|_{k,\alpha} \leq 2 \cdot c_k \quad \implies \quad \|w_{n+1}\|_{k,\alpha} \leq 2 \cdot c_k$$

for any n . Since $w_0 = 0$, we get ❷ for k . □

H Proof of Nash's lemma

Proof of 18. Recall that we have global periodic coordinates on $\mathbb{T}^n$.

Choose a metric tensor h on Ω . Since all $\partial_i f$, $\partial_{ij} f$ are linearly independent, the following identities uniquely define a vector field w on $\mathbb{T}^2 \Omega$ that is spanned by $\partial_i f$, $\partial_{ij} f$

$$\langle \partial_i f, w \rangle = 0, \quad \langle \partial_{ij} f, w \rangle = h_{ij}$$

for all i and j .

Note that

$$\begin{aligned} L_{ij}(w) &= 2 \cdot Q_{ij}(f, w) = \\ &= \langle \partial_i f, \partial_j w \rangle + \langle \partial_j f, \partial_i w \rangle = \\ &= \partial_j \langle \partial_i f, w \rangle + \partial_i \langle \partial_j f, w \rangle - 2 \cdot \langle \partial_{ij} f, w \rangle, \end{aligned}$$

it defines the required right inverse $M: h \mapsto w$. $\square$

Let $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ be a regular smooth map and $p \in \mathbb{T}^n$. Denote by T_p the n -dimensional space spanned by the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ at p . Denote by $T_p^\perp$ the normal complement of T_p in $\mathbb{R}^q$. Let us denote by $\mathbb{T}\mathbb{T}^n$ and $T_p^\perp \mathbb{T}^n$ the corresponding bundles over Ω . Further, denote by T_p^2 the space spanned by all first and second-order partial derivatives $\partial_i f$, $\partial_{ij} f$ at p , and let $N_p = T_p^2 \cap T_p^\perp$.

If f is free, then T_p^2 and N_p are respectively $(n + \frac{n \cdot (n+1)}{2})$ and $\frac{n \cdot (n+1)}{2}$ -dimensional at any point p . In this case we will denote by $\mathbb{T}^2 = T^2 \mathbb{T}^n$ and $N = N \mathbb{T}^n$ the corresponding bundles over $\mathbb{T}^n$.

Suppose $w = Mh$ as in the proof above. Since $\langle \partial_i f, w \rangle = 0$ for every i , we get the following.

22. Observation. *The inverse M in Nash's lemma (18) can be chosen so that $Mh(p) \in N_p$ for any $p \in \mathbb{T}^n$.*

The following exercise provides a key to proof of Günther's lemma.

23. Exercise. *Let $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ be a free map. Given a tangent vector field v on $\mathbb{T}^n$ and a metric tensor h on $\mathbb{T}^n$, show that there is a unique vector field w in $N \mathbb{T}^n$ such that*

$$2 \cdot Q(f, v + w) = h.$$

In other words, given v , there is a right inverse $M_v: h \mapsto v + w$ of L with w in $N \mathbb{T}^n$.

I Proof of Günther's lemma

Many symmetric bilinear operators $\tilde{Q}$ satisfy $L\tilde{Q} = Q$. In particular, one can take $\tilde{Q} = MQ$, where M is provided by Nash's lemma (18). However, this choice of $\tilde{Q}$ does not satisfy the required inequalities in Günther's lemma.

The actual construction of $\tilde{Q}$ relies on extra freedom provided by 23. It says that, given a tangent vector field v we may choose M so that $Mh - v \in N\mathbb{T}^n$.

Recall that we have globally defined partial derivatives $\partial_1, \dots, \partial_n$ on coordinates on $\mathbb{T}^n$. In particular, we can define the Laplacian

$$\Delta u = \sum_s \partial_{ss} u.$$

The next statement easily follows from the so-called Schauder estimates — a result in elliptic partial differential equations. It could be also obtained directly using Fourier analysis on torus; we omit its proof.

24. Proposition. *Let $k \geq 2$ be an integer and $0 < \alpha < 1$. Then*

$$(\Delta - 1): C^{k,\alpha}(\mathbb{T}^n) \rightarrow C^{k-2,\alpha}(\mathbb{T}^n).$$

is a bi-Lipschitz isomorphism. In particular, the linear operator

$$(\Delta - 1): u \mapsto \Delta u - u$$

has a Lipschitz inverse

$$\textcircled{1} \quad (\Delta - 1)^{-1}: C^{k-2,\alpha}(\mathbb{T}^n) \rightarrow C^{k,\alpha}(\mathbb{T}^n)$$

Note that the operator $(\Delta - 1)^{-1}$ smooths function — it swallows a $C^{k-2,\alpha}$ -function and spits a $C^{k,\alpha}$ -function. Also, $(\Delta - 1)^{-1}$ commutes with partial derivatives; that is,

$$(\Delta - 1)^{-1} \partial_i = \partial_i (\Delta - 1)^{-1}.$$

These two properties of $(\Delta - 1)^{-1}$ will be important in the following proof.

Proof of 19. Recall that $Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle$. Let us show that

$$\textcircled{2} \quad (1 - \Delta)Q_{ij}(w, w) = A_{ij}(w) + \partial_i A_j(w) + \partial_j A_i(w),$$

where each A_{ij} , A_i and A_j is a linear combination of the following quadratic forms

$$\textcircled{3} \quad w \mapsto \langle \partial_a w, \partial_b w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_c w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_{cd} w \rangle.$$

for all possible indexes a, b, c, d .

Indeed,

$$\begin{aligned}
(1 - \Delta)Q_{ij}(w, w) &= \langle \partial_i w, \partial_j w \rangle - \\
&\quad - \sum_s [2 \cdot \langle \partial_{is} w, \partial_{js} w \rangle + \langle \partial_{iss} w, \partial_j w \rangle + \langle \partial_i w, \partial_{jss} w \rangle] = \\
&= \langle \partial_i w, \partial_j w \rangle - 2 \cdot \sum_s \langle \partial_{is} w, \partial_{js} w \rangle + 2 \cdot \langle \Delta w, \partial_{ij} w \rangle - \\
&\quad - \partial_i \langle \Delta w, \partial_j w \rangle - \partial_j \langle \Delta w, \partial_i w \rangle.
\end{aligned}$$

Whence **2** follows.

Choose $A = A_i$ or $A = A_{ij}$ for some i and j . Note that

$$\begin{aligned}
&\|A(w)\|_{0,\alpha} \leq c_2 \cdot \|w\|_{2,\alpha}^2, \\
&\|A(w)\|_{k-2,\alpha} \leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha}^2 + c_k \cdot \|w\|_{k-1,\alpha}^2.
\end{aligned}$$

for some fixed constants $c_2, c_3, \dots$. Indeed, it is sufficient to check inequalities **4** for each form in **3**, and the latter is straightforward.

Applying 24, we get

$$\begin{aligned}
&\|(1 - \Delta)^{-1}A(w)\|_{2,\alpha} \leq c_2 \cdot \|w\|_{2,\alpha}^2, \\
&\|(1 - \Delta)^{-1}A(w)\|_{k,\alpha} \leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha} + c_k \cdot \|w\|_{k-1,\alpha}^2;
\end{aligned}$$

the constants $c_2, c_3, \dots$ might be different from those above, but each of them depends only on f .

Further, recall that

$$L_{ij}(w) = -2 \cdot \langle \partial_{ij} f, w \rangle + \partial_i \langle \partial_j f, w \rangle + \partial_j \langle \partial_i f, w \rangle;$$

see Section H. Let us define $\tilde{Q}$ as a linear combination of $\partial_i f$ and $\partial_{ij} f$ for all $i \leq j$ such that

$$\begin{aligned}
-2 \cdot \langle \partial_{ij} f, \tilde{Q}(w, w) \rangle &= (\Delta - 1)^{-1} A_{ij}(w), \\
\langle \partial_i f, \tilde{Q}(w, w) \rangle &= (\Delta - 1)^{-1} A_i(w)
\end{aligned}$$

for all i and j . Since f is free, $\tilde{Q}$ is well defined. Combining **6**, **7**, and **2**, we get

$$L\tilde{Q} = Q.$$

Let $\{e_i, e_{ij}\}_{i \leq j}$ be the dual frame to $\{\partial_i f, \partial_{ij} f\}_{i \leq j}$ in T^2 . Note that

$$\tilde{Q}(w, w) = \sum_i (\Delta - 1)^{-1} A_i(w) \cdot e_i - \frac{1}{2} \cdot \sum_{i \leq j} A_{ij}(w) \cdot e_{ij}.$$

Since the maps $e_i, e_{ij}: \mathbb{T}^n \rightarrow \mathbb{R}^q$ are smooth, **5** and 17 imply the last inequality in the lemma for each integer $k \geq 3$. $\square$

J Nash's twist

In this section we prove the following statement which can be regarded as an approximate version of our restricted theorem (7).

25. Proposition. *Let g be a Riemannian metric on $\mathbb{T}^n$. Then there is a metric tensor h on $\mathbb{T}^n$ and one-parameter family of smooth maps $w_t: \mathbb{T}^n \rightarrow \mathbb{R}^q$ for $t > 0$ such that*

$$Q(w_t, w_t) = g + t \cdot h$$

for any t .

Since $Q(w_t, w_t) \rightarrow g$ as $t \rightarrow 0$, it might be tempting to pass to the limit of w_t as $t \rightarrow 0$. However, as you will see, for the constructed maps w_t this limit is constant; so it does not solve 7.

The main player in the proof of the proposition is the so-called Nash's twist which are about to define.

Let φ and f be smooth functions on a smooth manifold Ω . Given $r > 0$, denote by $\mathbb{S}_r^1$ the circle of radius r in $\mathbb{R}^2$. The Nash's twist $F: \Omega \rightarrow \mathbb{R}^2$ of the triple (r, φ, f) is the following composition

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell_r} \mathbb{S}_r^1 \xrightarrow{\times \varphi} \mathbb{R}^2,$$

where $\ell_r: \mathbb{R} \rightarrow \mathbb{S}_r^1$ is a length-preserving covering map, say

$$\ell_r(x) = (r \cdot \cos \frac{x}{r}, r \cdot \sin \frac{x}{r}),$$

and $\times \varphi$ denotes the multiplication by φ ; so

$$F(x) := \varphi(x) \cdot (\ell_r \circ f(x)).$$

Recall that $(df)^2$ denotes the metric tensor induced by defined by $f: \Omega \rightarrow \mathbb{R}$; equivalently,

$$(df)^2(x, y) := df(x) \cdot df(y) = (xf) \cdot (yf).$$

26. Claim. *The metric tensor*

$$\varphi^2 \cdot (df)^2 + r^2 \cdot (d\varphi)^2$$

is induced by Nash's twist F for (r, φ, f) .

Computations. Suppose that x is a tangent vector on Ω , then

$$\begin{aligned} xF &= x(\varphi \cdot \ell_r \circ f) = \\ &= (x\varphi) \cdot (\ell_r \circ f) + \varphi \cdot (\ell'_r \circ f) \cdot (xf). \end{aligned}$$

Observe that $|\ell_r| = r$, $|\ell'_r| = 1$, and $\ell_r \perp \ell'_r$. Therefore,

$$\langle \mathbf{x}F, \mathbf{y}F \rangle = r^2 \cdot (\mathbf{x}\varphi) \cdot (\mathbf{y}\varphi) + \varphi^2 \cdot (\mathbf{x}f) \cdot (\mathbf{y}f). \quad \square$$

Let $w: \Omega \rightarrow \mathbb{R}^q$ be a smooth map defined on a smooth manifold Ω . Suppose that $w_1, \dots, w_q$ are coordinate functions of w , then the induced metric tensor g can be written as

$$g = Q(v, w) = (dw_1)^2 + \dots + (dw_q)^2.$$

In order to prove 7, we need to find functions $w_1, \dots, w_q: \Omega \rightarrow \mathbb{R}$ that solves the above equation for a given Riemannian metric. The following weaker form of 7 will play a key role in the proof of 25.

27. Proposition. *Let (Ω, g) be a compact n -dimensional Riemannian manifold. Then there are smooth functions $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ on Ω such that*

$$\textcircled{1} \quad g = (\varphi_1)^2 \cdot (df_1)^2 + \dots + (\varphi_q)^2 \cdot (df_q)^2.$$

Proof. The metric tensor g can be written in local cart $(x_1, \dots, x_n)$ as

$$\textcircled{2} \quad g = \sum_{i,j} g_{ij} \cdot dx_i \cdot dx_j,$$

where $g_{ij} = g_{ji}$ are smooth functions of $(x_1, \dots, x_n)$.

Since g is Riemannian, we can choose chart so that ∂_i are orthonormal at a given point p ; in other words, $g_{ii} = 1$ and $g_{ij} = 0$ at p for all $i \neq j$. Since g_{ij} are smooth, given $\varepsilon > 0$, we can choose a neighborhood $U \ni p$ such that

$$\textcircled{3} \quad g_{ii} \leq 1 \pm \varepsilon \quad \text{and} \quad g_{ij} \leq \pm \varepsilon$$

for all $i \neq j$ and any point in U .

Observe that

$$\pm dx_i \cdot dx_j = \frac{1}{2} \cdot (dx_i \pm dx_j)^2 - \frac{1}{2} \cdot (dx_i)^2 - \frac{1}{2} \cdot (dx_j)^2.$$

Plug this formula in $\textcircled{2}$ and take $\textcircled{3}$ into account, assuming that ε is small. We get that g is a linear combination with positive coefficients of the metric tensors $(dx_i)^2$ and $(dx_i \pm dx_j)^2$ for all i and j . In other words, we can take the functions x_i and $x_i \pm x_j$ for all i and j as our functions $f_1, \dots, f_q$ and find $\varphi_1, \dots, \varphi_q$ such that $\textcircled{1}$ holds in a neighborhood of p .

Applying a partition-of-unity argument, we get the global statement. $\square$

In the presented proof, it is sufficient to take $2 \cdot n^2$ functions in each chart; so, if Ω is covered by m charts, then one can take $q = 2 \cdot m \cdot n^2$ functions. However, as it is stated in the following exercise, one may take q independent of m .

28. Exercise. *Show that q in 27 can be bounded explicitly in terms of n ; say, we can assume that $q = 100 \cdot n^{100}$.*

Proof of 25. Suppose that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by 27. Let $r = \sqrt{t}$; denote by F_i the Nash's twist for the triple (r, φ_i, f_i) . By 26,

$$w_t = F_1 \oplus \dots \oplus F_q$$

is the needed map with $h = (d\varphi_1)^2 + \dots + (d\varphi_q)^2$. $\square$

K Proof assembling

Proof of 7. Choose a free map $f_1: \mathbb{T}^n \rightarrow \mathbb{R}^q$; it exists by 14. Suppose g_1 is the metric tensor induced by f_1 . We can assume that $g_1 < g$; that is, $g_1(x, x) < g(x, x)$ for any tangent vector $x \neq 0$. In particular, $g_2 = g - g_1$ is a Riemannian metric.

Applying ??, we get a metric tensor h on $\mathbb{T}^n$ such that for any $t > 0$ there is a smooth map $f_2: \mathbb{T}^n \rightarrow \mathbb{R}^{q_2}$ that induces $g_2 + t \cdot h$.

By the perturbation lemma, for any small $t > 0$ there is a smooth map $w: \mathbb{T}^n \rightarrow \mathbb{R}^{q_1}$ such that

$$Q(f_1 + w, f_1 + w) = g_1 - t \cdot h.$$

Finally by 5, the metric tensor

$$g = g_1 - t \cdot h + g_2 + t \cdot h$$

is induced by the map $(f_1 + w) \oplus f_2$. $\square$

L A pseudoeuclidean degression

Let us denote by $\mathbb{R}^r \ominus \mathbb{R}^s$ the pseudoeuclidean space with signature (r, s) ; that is, the space $\mathbb{R}^{r+s}$ with scalar product defined by

$$\langle x, y \rangle = x_1 \cdot y_1 + \dots + x_s \cdot y_s - x_{s+1} \cdot y_{s+1} - \dots - x_{s+r} \cdot y_{s+r},$$

where x_i and y_i denote the coordinates of vectors x and y in $\mathbb{R}^{r+s}$.

29. Problem. *Show that any metric g on a smooth manifold Ω is induced by a smooth embedding $\Omega \rightarrow \mathbb{R}^r \ominus \mathbb{R}^s$ for large r and s .*

Solution based on Nash's theorem. Note that $g = g_1 - g_2$ for two Riemannian metrics g_1 and g_2 on Ω . By Nash's theorem, we can find two smooth embeddings f_1 and f_2 of Ω to $\mathbb{R}^q$ with induced metrics g_1 and g_2 respectively. Then g is induced by the map

$$f_1 \ominus f_2: p \mapsto (f_1(p), f_2(p)) \in \mathbb{R}^q \ominus \mathbb{R}^q.$$

□

Now we will show that the problem can be solved using only Nash's twist.

Solution based on Nash's twist. Assume g is Riemannian and $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by 27. Let F_i be Nash's twist for the triple $(1, \varphi_i, f_i)$, and

$$\begin{aligned} h &= (d\varphi_1)^2 + \dots + (d\varphi_q)^2, \\ \mathbf{F} &= F_1 \oplus \dots \oplus F_q: \Omega \rightarrow \mathbb{R}^{2 \cdot q}, \\ \varphi &= \varphi_1 \oplus \dots \oplus \varphi_q: \Omega \rightarrow \mathbb{R}^q. \end{aligned}$$

According to 26, $\mathbf{F}$ induces $g + h$. Note that φ induces h . Therefore $S = \mathbf{F} \ominus \varphi: \Omega \rightarrow \mathbb{R}^{2 \cdot q} \ominus \mathbb{R}^q$ induces g .

To solve the general case, present the metric as a difference of two Riemannian metrics $g = g_1 - g_2$. From above, there is a smooth embeddings S_1 and S_2 into pseudoeuclidean spaces that induces g_1 and g_2 respectively. It remains to observe that $S_1 \ominus S_2$ induces g . □

30. Exercise. Let f be a smooth isometric immersion of a Riemannian manifold (Ω, g) into the sphere $\mathbb{S}^{q-1} \subset \mathbb{R}^q$ and let $\varphi: \Omega \rightarrow \mathbb{R}$ is a smooth function. Show that $\varphi^2 \cdot g$ is induced by the map $\varphi \cdot f \ominus \varphi$.

M Remarks

Let us state another closely related result that shows a huge difference between C^1 and C^2 isometric embeddings. For example, it implies that the unit sphere admits C^1 length-preserving embedding into an arbitrarily small ball in Euclidean 3-space. There is no such C^2 -embedding since Gauss curvature of the unit sphere is 1, but at an extremal point it must be at least $\frac{1}{r^2}$, where r is the radius of the ball.

31. Nash–Kuiper theorem. Let (Ω, g) be a n -dimensional Riemannian manifold and $f: (\Omega, g) \rightarrow \mathbb{R}^q$ be a short smooth regular map. Suppose that $q \geq n + 1$. Then for any $\varepsilon > 0$ there is an C^1 -smooth length-preserving maps $f_\varepsilon: (\Omega, g) \rightarrow \mathbb{R}^q$ that is ε close to f ; that is, $|f_\varepsilon(x) - f(x)| < \varepsilon$ for any $x \in \Omega$.

Moreover if f is an embedding then we can assume that so is f_ε .

It was originally proved by John Nash [5] with the condition $q \geq n+2$ and improved to $q \geq n+1$ by Nicolaas Kuiper [4]. The original proof uses Nash's twist in a different way. Both papers are reader-friendly. One may also start with lectures by Allan Yashinski and the author [7] where related results were obtained using an alternative approach.

The discussed result formed a part of foundations of the so-called homotopy principle, or h-principle; an excellent introduction is given in the book by Yakov Eliashberg and Nikolai Mishachev [1].

Many related questions are open. For example, it is unknown if a neighborhood of any point in 2-dimensional Riemannian manifold admits a smooth length-preserving embedding into $\mathbb{R}^3$.

The compactness condition can be removed; we will give a hint how to do it. Moreover the value q can be estimated in terms $n = \dim \Omega$. We will give a hint how to get a ruf bound, say $q = 100 \cdot n^3$. The best known estimate so far is

$$q = \max\left\{\frac{n \cdot (n+5)}{2}, \frac{n \cdot (n+3)}{2} + 5\right\}.$$

A Semisolutions

2. Consider square of a parallel 1-form α on $\mathbb{T}^2$ with kernel in irrational direction.

4. The only-if part is trivial.

Applying the length-preserving property to all arcs of curves in the coordinate directions, show that the equality

$$g_{ii} = \langle \partial_i w, \partial_i w \rangle$$

holds almost everywhere, and therefore everywhere.

Observe that

$$2 \cdot g_{ij} = g(\partial_i + \partial_j, \partial_i + \partial_j) - g(\partial_i, \partial_i) - g(\partial_j, \partial_j).$$

Using this formula with the argument above for curves in the direction ∂_i , ∂_j and $\partial_i + \partial_j$, show that the equality

$$g_{ii} = \langle \partial_i w, \partial_j w \rangle$$

holds almost everywhere, and therefore everywhere.

5+6. Spell the definitions.

9. Applying the definition of submanifold, we can cover Ω by a rectangular charts where Ω is written as the coordinate subspace of the first $k = \dim \Omega$ coordinates. Note that in each chart we can present the metric on Ω as a restriction of a Riemannian metric on the whole chart.

Extend this collection of charts to an atlas of $\mathbb{T}^n$ such that the images of the remaining charts do not intersect Ω . Equip these new charts with a constant Riemannian metric.

Choose a partition of unity subordinate to the covering by charts and use it to mix the metrics on the charts into a metric on $\mathbb{T}^n$.

10. Let $\iota: \Omega \rightarrow \mathbb{R}^q$ be an embedding provided by the Whitney theorem; let us think that $\mathbb{R}^q$ is an affine subspace $x_{q+1} = 1$ in $\mathbb{R}^{q+1}$. The needed embedding can be found among maps $x \mapsto \varphi \cdot \iota$, where $\varphi: \Omega \rightarrow \mathbb{R}$ is a smooth positive function.

Try to write down the necessary conditions and convince yourself that there is such function φ .

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Bibliography

- [1] Y. Eliashberg and N. Mishachev, *Introduction to the h-principle*. Vol. 48. Graduate Studies in Mathematics. 2002.
- [2] M. Günther, “On the perturbation problem associated to isometric embeddings of Riemannian manifolds”. *Ann. Global Anal. Geom.* 7.1 (1989), 69–77.
- [3] R. Howard. *Notes on Günther’s Method and the Local Version of the Nash Isometric Embedding Theorem*. URL: <https://people.math.sc.edu/howard/Notes/nash.pdf>.
- [4] N. Kuiper, “On C^1 -isometric imbeddings. I, II”. *Nederl. Akad. Wetensch. Proc. Ser. A.* **58** = *Indag. Math.* 17 (1955), 545–556, 683–689.
- [5] J. Nash, “ C^1 isometric imbeddings”. *Ann. of Math. (2)* 60 (1954), 383–396.
- [6] J. Nash, “The imbedding problem for Riemannian manifolds”. *Ann. of Math. (2)* 63 (1956), 20–63.
- [7] A. Petrunin and A. Yashinski, “Piecewise isometric mappings”. *Algebra i Analiz* 27.1 (2015), 218–247.
- [8] T. Tao. *Notes on the Nash embedding theorem*. URL: <https://terrytao.wordpress.com/2016/05/11/>.
- [9] D. Yang, *Gunther’s proof of Nash’s isometric embedding theorem*. 1998.